Zenith Echo

ZECATHON 6.0 · INCLUSIVE DIGITAL BANKING

Bank with your voice. Keep your PIN.

A hands-free assistant inside the Zenith app for customers who are blind, have low vision, or find reading hard.

zenith-echo.onrender.com

THE TEAM

Daniel Toluwani Olorunda
Team lead and product

Lemeh Godwin Chisom
Frontend and interface

Anjorin Olamide
Voice, language and backend

Inioluwa Aroyewun
Integration and testing

“Hey Ziva, send five thousand to my mechanic.”

A stranger offered to help a visually impaired widow at the ATM.

She learned the PIN, swapped the card for an expired one, and withdrew over ₦2 million. The widow only found out from the debit alerts.

When you can’t see the screen, “help” is the risk.

Ripples Nigeria
19 Dec 2024

IT IS NOT ONE PERSON

5.1M

Nigerian adults live with a disability, about 8% of adult consumers.

40M

adults are still outside formal finance. Formal inclusion is 64%.

38%

of adults cannot read comfortably, yet banking apps are built on text.

NBS via BusinessDay (Apr 2025) · EFInA A2F 2023 via The Guardian (Dec 2024) · UNESCO via BusinessDay

Echo drives the real app. It stops at the PIN.

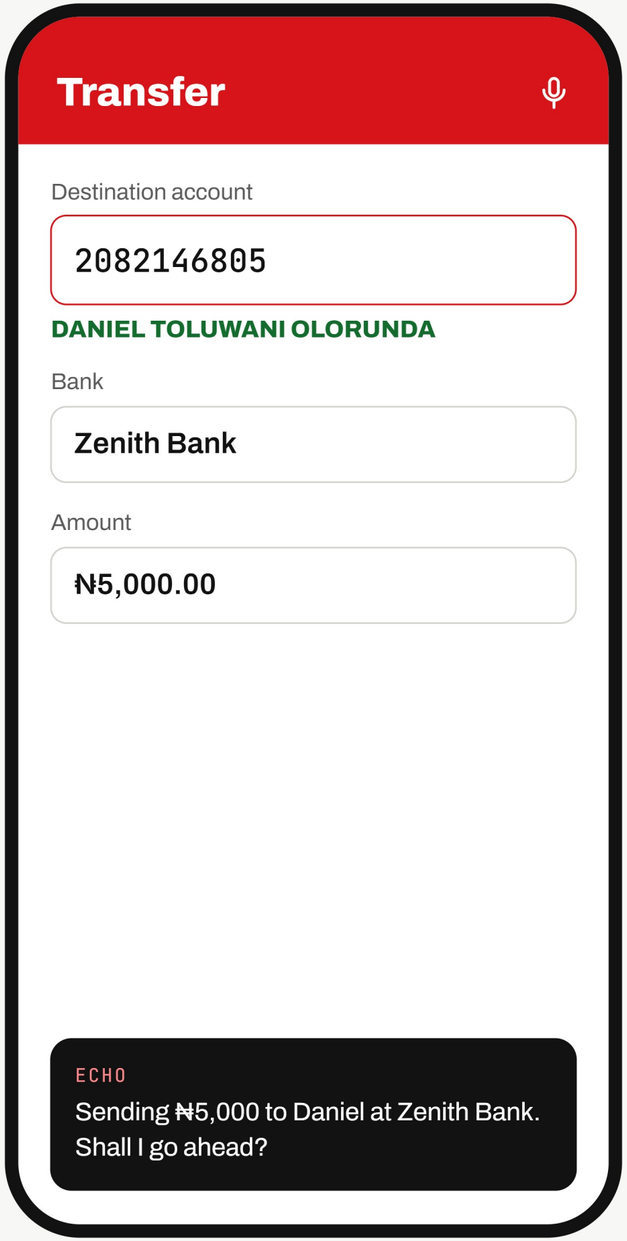
Not a chatbot beside the app. The customer speaks, Echo opens the real Transfer, Airtime, Data or Electricity screen and fills every field. The customer types the PIN, so no helper ever needs it.

“Hey Ziva” on every screen
Speak over Echo to interrupt it.

English, Pidgin, Yoruba
“Abeg send 5k give my mama” works.

Whisper privacy
Balances stay silent without headphones.

Touch and read
Tap to hear, double tap to press.



Voice in. Money decisions stay on the server.

PHONE

Zenith app

React, TypeScript. Wake word, barge-in, touch and read, voice fills the real forms.

⇄

SERVER

Echo API

Node, Express. Intent in 3 languages, a guarded checklist that never assumes a detail.

⇄

SPEECH

ElevenLabs

Multilingual voice and transcription. Key held on the server. Private details redacted.

Voice can fill a form. It can never approve a payment.

3 tries
Hashed PIN, locks after three wrong attempts.

2 min
Single-use approval for one transaction.

₦50k
Above this, PIN only. No fingerprint.

NUBAN
CBN check digit catches a misheard number.

Never
A spoken PIN is ignored, with a warning.

Millions are already Zenith customers.

33M

Zenith customers at end of 2023, growing mostly on digital.

~2.6M

of them may live with a disability, if the national 8% holds. Our estimate.

Jan 2024

The Disability Act's five-year accessibility grace period ended.

Nairametrics (May 2024) · NBS via BusinessDay (Apr 2025) · NAN (Sep 2023) · CITAB via Inclusive News (Nov 2024)

Why it's different from what blind customers use today

	PIN STAYS PRIVATE	NO READING	NO PRECISE TAPS	PIDGIN, YORUBA
A relative or agent	**No**	Yes	Yes	Yes
Phone screen reader	Yes	Yes	**No**	**No**
USSD codes	Yes	**No**	**No**	**No**
Zenith Echo	**Yes**	**Yes**	**Yes**	**Yes**

Blind customers have reported being refused ATM cards and asked to sign indemnity forms. Echo gives Zenith a way to serve them independently and safely, in the app it already runs.

06 · WITH THREE MORE MONTHS

Validate with blind customers. Then ship it inside Zenith.

MONTH 1 · VALIDATE

Test with blind and low-vision customers through the Nigeria Association of the Blind and CITAB.
Measure completion, time and errors per transfer.

MONTH 2 · INTEGRATE

Native Android, so “Hey Ziva” works with the screen off.
Connect to Zenith sandbox APIs.
A Nigerian voice for Echo.

MONTH 3 · PILOT

Opt-in rollout to a group of real customers.
Fraud monitoring and limits on voice payments.
Hausa and Igbo next.

Pilot goal: a blind customer completes a transfer alone, with no one learning their PIN.

Our ask: a Zenith pilot

Eyes closed. One transfer.

YOU	**“Hey Ziva, I want to send money.”**
ECHO	Opens Transfer. Asks for the account.
YOU	**“2 0 8 2 1 4 6 8 0 5, Zenith.”**
ECHO	Checks it, reads the name. Asks how much.
YOU	**“Five thousand naira.” “Yes.”**
ECHO	Opens the PIN keypad. You type it.

Sent. Receipt read aloud. Thank you

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Demo accounts only. No real money moves.